LABORATORY · Lab · 26 hours

Experiment 2: Effect of Addition of Poles and Zeros on Stability



Control Systems Laboratory (22EEG53)

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Learning objectives

- Observe how an added open-loop pole or zero changes the root locus
- Observe the effect on the step response
- Relate the result to stability and relative stability

Syllabus context

Laboratory component of 22EEG53 Control Systems (0:0:2 hours per week). CIE: record and observation 25, viva-voce 5, lab test 20 marks.

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Aim and requirements

Aim: to study the effect of adding poles and zeros to a system on its stability using MATLAB.

Software: MATLAB with the Control System Toolbox (Simulink where required).

Hardware: Personal computer.

Base system: G(s) = K/[s(s + 2)] with unity feedback.

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Theory

Adding a pole

Bends the locus to the right: reduces stability, lower damping, may cause instability at high gain.

Adding a zero

Bends the locus to the left: increases stability and damping; adds overshoot via derivative-like action.

Closed-loop effect

Poles ⇒ slower response; zeros ⇒ faster response with more overshoot.

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Procedure

1

Base system

Plot root locus and step response of G = 1/(s(s+2)).

2

Add pole

Multiply by 1/(s + p): e.g. p = 4.

3

Add zero

Multiply by (s + z): e.g. z = 4.

4

Compare

Plot loci and closed-loop steps; note stability limits.

Observe the K for which the poles cross the jω axis.

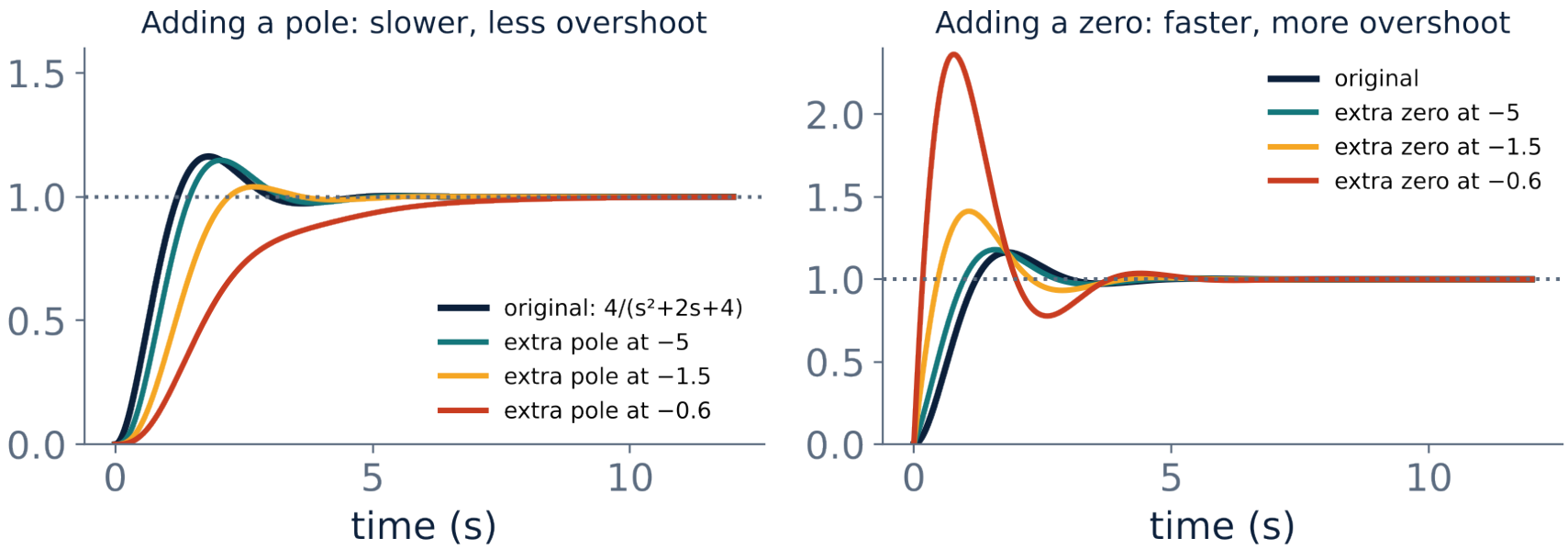
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MATLAB program

```
G  = tf(1, [1 2 0]);              % base
G1 = tf(1, conv([1 2 0],[1 4]));     % extra pole at -4
G2 = tf([1 4], [1 2 0]);             % extra zero at -4
subplot(2,3,1), rlocus(G),  title('base')
subplot(2,3,2), rlocus(G1), title('+ pole')
subplot(2,3,3), rlocus(G2), title('+ zero')
subplot(2,1,2), hold on
step(feedback(G,1)), step(feedback(G1,1)), step(feedback(G2,1))
legend('base','+pole','+zero'); grid on
```

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Expected step responses



Left: extra pole slows the response. Right: extra zero speeds it up and increases overshoot.

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Observation table

Case	Locus bends	Stability limit (K)	Overshoot	Settling time
Base K/[s(s+2)]	Vertical	Stable for all K		
+ pole at −4	Right	Finite K		
+ zero at −4	Left	Stable for all K		

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Key takeaways

- Adding a pole moves the locus right: less stable
- Adding a zero moves the locus left: more stable
- Poles slow the response; zeros speed it and increase overshoot
- MATLAB rlocus and step show both effects

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Practice questions

Q1. How does adding a pole change the stability of K/[s(s+2)]?

Q2. Which compensator adds a zero to the loop?

Q3. Write the MATLAB commands used.

Q4. Why does a zero increase overshoot?